

Differential Geometry

Exercise 6

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6.1 Problem

Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$(y_1, y_2) = f(x_1, x_2) = (x_1^3 + x_1x_2, x_2).$$

- (1) Graph y_1 as a function of x_1 and x_2 .
- (2) What are the critical points of f ?
- (3) What are the critical values of f ?
- (4) A line in $\mathbb{R}^2$ is a submanifold $\ell \subseteq \mathbb{R}^2$ determined by an equation of the form

$$ay_1 + by_2 = c$$

for constants $a, b, c \in \mathbb{R}$. Which lines are transverse to f ?

- (5) Draw $f^{-1}\ell$ and determine whether it is a submanifold of $\mathbb{R}^2$ for the values

$$(a, b, c) = (1, 0, -1), (1, 0, 0), (1, 0, 1), (1, -1, 1).$$

- (1) If we keep x_2 constant, the map $y_1 = x_1^3 + x_1x_2$ has derivative $3x_1^2 + x_2$, which can be zero only when $x_2 < 0$. So the graph of f at x_2 is $x_1^3 + x_1x_2$, which is just x_1^3 when $x_2 = 0$, and then as x_2 gets more negative it becomes a more and more extreme N shape, and as x_2 gets more positive it becomes closer to a line. I have no artistic skills, and was not able to draw the graph in any comprehensible way.
- (2) We compute f 's Jacobian: let $p = (x_1, x_2)$ then

$$df_p = \begin{pmatrix} 3x_1^2 + x_2 & x_1 \\ 0 & 1 \end{pmatrix}.$$

Since the domain and codomain of f have the same dimension, df_p is surjective iff it is injective iff it is bijective, which is equivalent to df_p having nonzero determinant. The determinant of df_p is easy to compute:

$$\det(df_p) = 3x_1^2 + x_2.$$

Now p is a critical point iff df_p is not surjective, so iff this determinant is zero. That is,

$$p \text{ is critical} \iff 3x_1^2 + x_2 = 0.$$

So f 's critical points are of the form $(t, -3t^2)$ for $t \in \mathbb{R}$.

- (3) f 's critical values are just the values it takes on its critical points. For a critical point $p = (t, -3t^2)$,

$$f(p) = (t^3 - 3t^3, -3t^2) = (-2t^3, -3t^2).$$

So this is the set of f 's critical values: all points in $\mathbb{R}^2$ of the form $(2t^3, 3t^2)$.

- (4) Recall that $f: N \rightarrow M$ is transverse to $S \subseteq M$ iff $df_p(T_p N) + T_p S = T_p M$ for all $p \in f^{-1}S$. If p is a regular point of f , this obviously holds. So we are interested in lines which contain a critical value of f .

First let us compute $T_q\ell$. Note that ℓ is the preimage of c under the smooth map

$$g: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad g(y_1, y_2) = ay_1 + by_2.$$

Thus $T_q\ell \subseteq T_q\mathbb{R}^2 = \mathbb{R}^2$ is just the kernel of dg_q . g 's differential is

$$dg = (a, b),$$

whose kernel is then

$$\ker dg = \ker(a, b) = \text{span}(b, -a)^\top.$$

Thus for $q \in \ell$,

$$T_q\ell = \text{span}(b, -a) \subseteq \mathbb{R}^2.$$

Now recall that we are interested only in points of the form $q = fp$ where $p = (x_1, x_2)$ is a critical point of f . Let $p = (t, -3t^2)$ be a critical point of f , then

$$df_p(T_p\mathbb{R}^2) = \text{im} \begin{pmatrix} 0 & t \\ 0 & 1 \end{pmatrix} = \text{span}(t, 1)^\top.$$

So we want to now find for which lines ℓ we have that for all p critical and $fp \in \ell$, $df_p(T_p\mathbb{R}^2) + T_{fp}\ell = T_{fp}\mathbb{R}^2 = \mathbb{R}^2$. Let $p = (t, -3t^2)$, then $fp = (-2t^3, -3t^2)$ is in ℓ when

$$-2at^3 - 3bt^2 = c.$$

And in such a case, $df_p(\mathbb{R}^2) + T_{fp}\ell = \mathbb{R}^2$ iff $(b, -a)$ and $(t, 1)$ together span $\mathbb{R}^2$, which is iff they form a basis iff

$$\det \begin{pmatrix} t & b \\ 1 & -a \end{pmatrix} = -at - b \neq 0.$$

So f is not transverse to ℓ if there is a $t \in \mathbb{R}$ such that

$$-2at^3 - 3bt^2 = c, \quad at + b = 0.$$

If $a \neq 0$ then we must have $t = -b/a$ and

$$2at^3 + 3bt^2 = 2a\left(-\frac{b}{a}\right)^3 + 3b\left(-\frac{b}{a}\right)^2 = \frac{b^3}{a^2},$$

and thus $c = -b^3/a^2$.

If $a = 0$ then we must have $b = 0$, in which case either $c = 0$ so $\ell = \mathbb{R}^2$ to which f is clearly transverse, or $c \neq 0$ so $\ell = \emptyset$ which again f is also transverse (vacuously; its preimage is empty).

So f is not transverse to lines of the form

$$ay_1 + by_2 = -\frac{b^3}{a^2}.$$

- (5) If f is transverse to S , then $f^{-1}S$ is a submanifold. For $(a, b, c) = (1, 0, -1), (1, 0, 1)$, f is transverse to ℓ so $f^{-1}\ell$ is a submanifold. We also compute what $f^{-1}\ell$ is in both of these cases.

For $(a, b, c) = (1, 0, -1)$, $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 = -1$. Since $x_1 = 0$ gives no solutions, this is iff

$$x_2 = -\frac{x_1^3 + 1}{x_1},$$

so $f^{-1}\ell$ is the set of all points of the form $\left(t, -\frac{t^3+1}{t}\right)$, i.e. it is the graph of $-\frac{t^3+1}{t}$.

Similarly for $(a, b, c) = (1, 0, 1)$, $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 = 1$ iff $x_2 = \frac{1-x_1^3}{x_1}$. So $f^{-1}\ell$ is the graph of $\frac{1-t^3}{t}$.

For $(a, b, c) = (1, 0, 0)$ we showed that f is not transverse to ℓ . Note that $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 = 0$, and the set of such points does not form a submanifold. Indeed,

$$x_1^3 + x_1x_2 = 0 \iff x_1 = 0 \text{ or } x_2 = -x_1^2,$$

so $f^{-1}\ell$ is just the union of the parabola $y = -x^2$ with the y -axis $x = 0$. This is not even a topological manifold, since it is not locally Euclidean around the origin: removing it creates 4 disconnected components, but removing a single point from one-dimensional Euclidean space creates only two. So $f^{-1}\ell$ is not a submanifold.

For $(a, b, c) = (1, -1, 1)$ we have that $f(x_1, x_2) \in \ell$ iff $x_1^3 + x_1x_2 - x_2 = 1$, and since (by polynomial division)

$$x_1^3 + x_1x_2 - x_2 - 1 = (x_1 - 1)(x_1^2 + x_1 + x_2 + 1),$$

$f^{-1}\ell$ is the union of the parabola $y = -x^2 - x - 1$ with the line $x = 1$; this is not a manifold for the same reason as before.

6.2 Problem

Show that the Levi-Civita connection on a pseudo-Riemannian manifold exists and is unique. Provide a full proof.

The derivation of Koszul's formula done in class does not change: the Levi-Civita connection must satisfy

$$\langle \nabla_X Y, Z \rangle = \frac{1}{2} (X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle - \langle Y, [X, Z] \rangle - \langle Z, [Y, X] \rangle + \langle X, [Z, Y] \rangle).$$

A pseudo-Riemannian metric is nondegenerate, so this uniquely defined $\nabla_X Y$. To show that this defines a symmetric metric connection, we focus on a coordinate frame given by a chart $(U, (x^i))$. The Gram matrix of a nondegenerate bilinear form is invertible, so the formula for the connection coefficients derived in class remains valid:

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}),$$

where $\partial_i = \partial/\partial x^i$. Note then that this defines a connection on $U \subseteq M$: the connection coefficients on a local frame define a connection. That is,

$$\nabla_X Y = (X(Y^k) + X^i Y^j \Gamma_{ij}^k) \partial_k$$

defines a connection on U .

We will show that this connection is symmetric and metric on U , and thus by uniqueness these definitions must coincide on the intersections of any charts, and thus gluing them together gives the desired Levi-Civita connection.

Notice that by g 's symmetry, $\Gamma_{ij}^k = \Gamma_{ji}^k$. Thus ∇ is symmetric, since

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k = \Gamma_{ji}^k \partial_k = \nabla_{\partial_j} \partial_i.$$

Then since the torsion tensor

$$\tau(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$$

is indeed a tensor, by linearity we see that since $\tau(\partial_i, \partial_j) = 0$ for all i, j , τ is identically zero. Thus ∇ is symmetric.

Now,

$$\nabla g(Y, Z, X) = X(g(Y, Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z)$$

is a tensor. Additivity in each component is easily seen, as well as multiplicativity in X . Multiplicativity in e.g. Y is easily shown,

$$\begin{aligned} \nabla g(fY, Z, X) &= X(g(fY, Z)) - g(\nabla_X fY, Z) - g(fY, \nabla_X Z) \\ &= (Xf)g(Y, Z) + fX(g(Y, Z)) - g((Xf)Y, Z) - fg(\nabla_X Y, Z) - fg(Y, \nabla_X Z) \\ &= fX(g(Y, Z)) - fg(\nabla_X Y, Z) - fg(Y, \nabla_X Z) \\ &= f\nabla g(Y, Z, X). \end{aligned}$$

Showing ∇ is compatible with g is the same as showing $\nabla g = 0$, and since this is a tensor it is sufficient to show this in our coordinate frame.

We have

$$g(\nabla_{\partial_i} \partial_j, \partial_k) = g(\Gamma_{ij}^\ell \partial_\ell, \partial_k) = \Gamma_{ij}^\ell g_{\ell k}.$$

By our formula for Γ_{ij}^k in this coordinate frame, this is equal to

$$= \frac{1}{2}(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}).$$

Thus

$$\begin{aligned} g(\nabla_{\partial_i} \partial_j, \partial_k) + g(\partial_j, \nabla_{\partial_i} \partial_k) &= \frac{1}{2}(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij} + \partial_i g_{kj} + \partial_k g_{ij} - \partial_j g_{ik}) \\ &= \partial_i g_{jk} = \partial_i g(\partial_j, \partial_k). \end{aligned}$$

Showing that ∇ is indeed compatible with g .

As explained, our formula for ∇ defines a unique symmetric metric connection on each coordinate frame. Because it is unique, the formulas must agree where the coordinate frames intersect, thus allowing us to glue them together to give a global symmetric metric connection.

6.3 Problem

Let $(M, g), (N, h)$ be Riemannian manifolds, and $\alpha: M \rightarrow N$ an isometric immersion. Let $\bar{\nabla}$ be the Levi-Civita connection on N , and $\tilde{\nabla}$ be its pullback along α . Consider the orthogonal projection $\Pi_q: T_{\alpha q} N \rightarrow d\alpha_q(T_q M)$ for $q \in M$, and let $P_q: T_{\alpha q} N \rightarrow T_q M$ denote its composition with the isomorphism $d\alpha_q(T_q M) \cong T_q M$.

For $X \in \mathfrak{X}(M)$ define $\hat{X} \in \mathfrak{X}(\alpha)$ by $\hat{X}_q = d\alpha_q(X_q)$.

- (1) Show there exists a unique $C^\infty(M)$ -linear map $P: \mathfrak{X}(\alpha) \rightarrow \mathfrak{X}(M)$ such that

$$P(X)_q = P_q(X_q)$$

for all $q \in M, X \in \mathfrak{X}(\alpha)$.

- (2) Define a connection ∇ on M by $\nabla X = P \circ \tilde{\nabla} \hat{X}$. Prove that ∇ is the Levi-Civita connection on M .

- (1) Uniqueness is clear as the formula given defines P . For $f^1, f^2 \in C^\infty(M)$ and $X_1, X_2 \in \mathfrak{X}(\alpha)$ we have, by linearity of P_q ,

$$P(f^i X_i)_q = P_q(f^i(q) X_i(q)) = f^i(q) P_q(X_i(q)) = (f^i P(X_i))_q,$$

so $P(f^i X_i) = f^i P(X_i)$ as desired.

Now we show that P maps into $\mathfrak{X}(M)$. By linearity it is sufficient to show this for local frames. Since α is an immersion, locally it has a coordinate representation $\bar{x} \mapsto (\bar{x}, \bar{0})$. Let $x^1, \dots, x^m$ be local coordinates for M around q and $y^1, \dots, y^n$ be local coordinates for N around $\alpha(q)$ giving this representation.

Then

$$d\alpha_q \left(\frac{\partial}{\partial x^i} \Big|_q \right) = \frac{\partial \alpha^j}{\partial x^i}(q) \frac{\partial}{\partial y^j} \Big|_{\alpha(q)},$$

and since under these coordinates α is the map $\bar{x} \mapsto (\bar{x}, \bar{0})$, its Jacobian is just $\partial \alpha^j / \partial x^i = \delta_{ij}$. Thus

$$d\alpha_q \left(\frac{\partial}{\partial x^i} \Big|_q \right) = \frac{\partial}{\partial y^i} \Big|_{\alpha q}.$$

And so $d\alpha_q(T_q M)$ is spanned by $\partial / \partial y^i|_{\alpha q}$ for $1 \leq i \leq m$. Let Π_p^N denote the orthogonal projection of $T_p N$ onto the subspace spanned by $\partial / \partial y^i|_p$ for $1 \leq i \leq m$. This is a smooth map, as projecting is just the application of Gram-Schmidt which is smooth, and $\Pi_q = \Pi_{\alpha q}^N$. Thus

$$P(X)_q = P_q(X_q) = (d\alpha_q)^{-1}(\Pi_{\alpha q}^N X_q),$$

which is smooth in q as a composition of smooth maps. Thus $P(X)$ is a smooth vector field, as desired.

(2) Recall that $\tilde{\nabla}$ is given by

$$\tilde{\nabla}_X(Y \circ \alpha)|_q = \bar{\nabla}_{\tilde{X}_q} Y|_{\alpha q}$$

for $X \in \mathfrak{X}(M)$, $Y \in \mathfrak{X}(N)$, $q \in M$.

We first show that ∇ is indeed a connection; we utilize the fact that $\tilde{\nabla}$ is a connection itself. For $C^\infty(M)$ -linearity in the first argument we have

$$\nabla_{f^i X_i} Y = P(\tilde{\nabla}_{f^i X_i} \hat{Y}) = P(f^i \tilde{\nabla}_{X_i} \hat{Y}) = f^i P(\tilde{\nabla}_{X_i} \hat{Y}) = f^i \nabla_{X_i} Y,$$

where the second-to-last equality is due to P being $C^\infty(M)$ -linear. Now notice that $\hat{\bullet}$ is also $C^\infty(M)$ -linear:

$$\widehat{f^i X_i}_q = d\alpha_q(f^i(q) X_i|_q) = f^i(q) d\alpha_q(X_i|_q) = (f^i \hat{X}_i)_q.$$

This allows us to show the rest of the properties of connections. For $\mathbb{R}$ -linearity in the second argument we have

$$\nabla_X c^i Y_i = P(\tilde{\nabla}_X c^i \hat{Y}_i) = c^i P(\tilde{\nabla}_X \hat{Y}_i) = c^i \nabla_X Y_i.$$

And for Leibniz,

$$\nabla_X f Y = P(\tilde{\nabla}_X f \hat{Y}) = P((Xf) \hat{Y} + f \tilde{\nabla}_X \hat{Y}) = (Xf) P(\hat{Y}) + f \nabla_X Y.$$

Now $P(\hat{Y}) = Y$, since $P(\hat{Y})_q = P_q(\hat{Y}_q) = P_q(d\alpha_q(Y_q))$, since $d\alpha_q(Y_q)$ is already in $d\alpha_q(T_q M)$ the orthogonal projection does not affect it, so this is just mapped to Y_q . Thus

$$\nabla_X f Y = (Xf)Y + f \nabla_X Y$$

as desired.

Now we show that ∇ is symmetric. It is sufficient to show this for local frames of our choice since the torsion tensor $(\nabla_X Y - \nabla_Y X - [X, Y])$ is a tensor, and thus linear. In particular we will use the coordinate frame $\partial / \partial x_i|_q$ from the first point, so

$$\widehat{\frac{\partial}{\partial x^i}}_q = d\alpha_q \left(\frac{\partial}{\partial x^i} \Big|_q \right) = \frac{\partial}{\partial y^i} \Big|_{\alpha(q)},$$

meaning

$$\widehat{\frac{\partial}{\partial x^i}} = \frac{\partial}{\partial y^i} \circ \alpha.$$

Then

$$\nabla_{\partial/\partial x^i} \frac{\partial}{\partial x^j} \Big|_q = P_q \left(\widetilde{\nabla}_{\partial/\partial x^i} \widehat{\frac{\partial}{\partial x^j}} \Big|_q \right).$$

Recall that $\widehat{\partial/\partial x^i} = \partial/\partial y^i \circ \alpha$, we have that

$$\widetilde{\nabla}_{\partial/\partial x^i} \frac{\partial}{\partial y^j} \circ \alpha \Big|_q = \overline{\nabla}_{\partial/\partial y^i|_{\alpha q}} \frac{\partial}{\partial y^j} \Big|_{\alpha q} = \overline{\nabla}_{\partial/\partial y^i} \frac{\partial}{\partial y^j} \Big|_{\alpha q}.$$

If we let Γ_{ij}^k be the Christoffel symbols of (N, h) , then

$$\nabla_{\partial/\partial x^i} \frac{\partial}{\partial x^j} \Big|_q = P_q \left(\Gamma_{ij}^k(\alpha q) \frac{\partial}{\partial y^k} \Big|_{\alpha q} \right) = \Gamma_{ij}^k(\alpha q) P_q \left(\frac{\partial}{\partial y^k} \Big|_{\alpha q} \right).$$

All in all this means that

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = (\Gamma_{ij}^k \circ \alpha) P \left(\frac{\partial}{\partial y^k} \circ \alpha \right).$$

Since $\overline{\nabla}$ is symmetric, $\Gamma_{ij}^k = \Gamma_{ji}^k$, and thus this shows

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^i}.$$

Since the Lie bracket of coordinate vector fields is zero, this gives the desired result: ∇ is symmetric.

Finally we must show that ∇ is compatible with M 's metric. Again this can be checked locally on a local frame of our choice (since ∇g is a tensor, as shown in our previous solution). We remain with our local coordinate frame as before, so we need to show

$$\frac{\partial}{\partial x^i} \left\langle \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g = \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g + \left\langle \frac{\partial}{\partial x^j}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k} \right\rangle_g,$$

where $\langle \bullet, \bullet \rangle_g$ is just $g(\bullet, \bullet)$.

We first focus on the left-hand side. Notice that for $i, j, k \leq m$,

$$\frac{\partial}{\partial y^i} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \Big|_{\alpha q} = \frac{\partial}{\partial y^i} \Big|_{\alpha q} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h = d\alpha_q \left(\frac{\partial}{\partial x^i} \Big|_q \right) \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h.$$

By definition this is just

$$= \frac{\partial}{\partial x^i} \Big|_q \left(\left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \circ \alpha \right).$$

Recall that $\partial/\partial y^i \circ \alpha = d\alpha \partial/\partial x^i$, so this is equal to

$$= \frac{\partial}{\partial x^i} \Big|_q \left\langle d\alpha \frac{\partial}{\partial x^j}, d\alpha \frac{\partial}{\partial x^k} \right\rangle_h,$$

and since α is an isometry, this is equal to

$$= \frac{\partial}{\partial x^i} \Big|_q \left\langle \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g.$$

Thus we have shown

$$\frac{\partial}{\partial x^i} \left\langle \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle = \frac{\partial}{\partial y^i} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle \circ \alpha.$$

Now we consider the right-hand side. Recall that we already computed $\nabla_{\partial/\partial x^i} \partial/\partial x^j$, and we apply that here:

$$\left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g = \left\langle (\Gamma_{ij}^\ell \circ \alpha) P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial x^k} \right\rangle_g = (\Gamma_{ij}^\ell \circ \alpha) \left\langle P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial x^k} \right\rangle_g.$$

Since α is an isometry, we have

$$\left\langle P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial x^k} \right\rangle_g = \left\langle d\alpha P \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial y^k} \circ \alpha \right\rangle_h.$$

By definition of P this is equal to

$$= \left\langle \Pi \left(\frac{\partial}{\partial y^\ell} \circ \alpha \right), \frac{\partial}{\partial y^k} \circ \alpha \right\rangle_h.$$

Now Π is the orthogonal projection onto the subspace spanned by $\partial/\partial y^\ell$ for $1 \leq \ell \leq m$, and thus does not affect this inner product:

$$= \left\langle \frac{\partial}{\partial y^\ell} \circ \alpha, \frac{\partial}{\partial y^k} \circ \alpha \right\rangle_h = \left\langle \frac{\partial}{\partial y^\ell}, \frac{\partial}{\partial y^k} \right\rangle_h \circ \alpha.$$

All in all we have

$$\left\langle \nabla_{\partial/\partial x^i} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g = \left(\Gamma_{ij}^\ell \left\langle \frac{\partial}{\partial y^\ell}, \frac{\partial}{\partial y^k} \right\rangle_h \right) \circ \alpha = \left\langle \nabla_{\partial/\partial y^i} \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \circ \alpha.$$

Putting all this together, ∇ 's compatibility with M 's metric follows from $\bar{\nabla}$'s:

$$\begin{aligned} \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right\rangle_g + \left\langle \frac{\partial}{\partial x^j}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k} \right\rangle_g &= \left(\left\langle \nabla_{\partial/\partial y^i} \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h + \left\langle \frac{\partial}{\partial y^j}, \nabla_{\partial/\partial y^i} \frac{\partial}{\partial y^k} \right\rangle_h \right) \circ \alpha \\ &= \left(\frac{\partial}{\partial y^i} \left\langle \frac{\partial}{\partial y^j}, \frac{\partial}{\partial y^k} \right\rangle_h \right) \circ \alpha \\ &= \frac{\partial}{\partial x^i} \left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right). \end{aligned}$$

Since the Levi-Civita connection is unique, ∇ must be M 's.

6.4 Problem

- (1) Do Carmo Riemannian Geometry, page 180 problem 3a, 3b. Consider $\mathbb{R}^{n+1}$ with the symmetric nondegenerate quadratic form

$$Q(x_0, \dots, x_n) = -x_0^2 + \sum_{i=1}^n x_i^2.$$

This induces a pseudo-Riemannian metric on $\mathbb{R}^{n+1}$, call the resulting pseudo-Riemannian mani-

fold L^{n+1} . For $r > 0$, let $k = -1/r^2$, and define by H_k^n the embedded submanifold of L^{n+1} given by

$$H_k^n = \{ \bar{x} \in L^{n+1} \mid x_0 > 0, Q(\bar{x}) = -r^2 \}.$$

Show that for $x \in H_k^n$, the vector $\eta = x/r$ is normal to the tangent space $T_x H_k^n$. Further show that $(\eta, \eta) = -1$ and it is possible to extend it to a basis $\eta, b_1, \dots, b_{n+1}$ of L^{n+1} such that $(b_i, b_j) = \delta_{ij}$ and $(b_i, b_0) = 0$. Deduce that the restriction of L^{n+1} 's metric to H_k^n is positive definite, and thus forms a Riemannian manifold.

- (2) Do Carmo Riemannian Geometry, page 184 problem 7a, 7b. Define a map $f: H_{-1}^n \rightarrow D^n$, where

$$D^n = \left\{ \bar{x} \in \mathbb{R}^n \mid x_0 = 0, \sum_{i=1}^n x_i^2 < 1 \right\},$$

as follows. For $p \in H_{-1}^n \subseteq L^{n+1}$, form a line from p to $p_0 = (-1, 0, \dots, 0)$, and let $f(p)$ be the intersection of this line with D^n . Let $f(\bar{x}) = (0, \bar{u})$, and show

$$x_i = \frac{2u_i}{1 - \sum_j u_j^2}, \quad i = 1, \dots, n,$$

$$x_0 = \frac{2}{1 - \sum_j u_j^2} - 1.$$

Now show that

$$-(dx_0)^2 + \sum_{i=1}^n (dx_i)^2 = \frac{4 \sum_{i=1}^n (du_i)^2}{\left(1 - \sum_j u_j^2\right)^2}.$$

Conclude that f^{-1} induces a Riemannian metric on D^n by

$$g_{ij} = \frac{4\delta_{ij}}{\left(1 - \sum_j u_j^2\right)^2}.$$

- (3) Let

$$\mathbb{R}_+^n = \{ \bar{x} \in \mathbb{R}^n \mid x_n > 0 \},$$

and define a metric on it by

$$g_{ij}(\bar{x}) = \frac{\delta_{ij}}{x_n^2}.$$

Let $p_0 = (0, \dots, -1) \in \mathbb{R}^n$ and define $f: D^n \rightarrow \mathbb{R}_+^n$ by

$$f(p) = 2 \frac{p - p_0}{|p - p_0|^2} + p_0.$$

Show that if we equip D with the metric from the previous point, f forms an isometry.

- (4) Equip D^n with a metric given by

$$g_{ij}(\bar{y}) = \frac{\delta_{ij}}{1 - |y|^2} + \frac{y_i y_j}{(1 - |y|^2)^2}.$$

Define $h: H_{-1}^n \rightarrow D^n$ by $y_i = x_i/x_0$. Show that h is an isometry.

- (1) So H_k^n is the level set of $-r^2$ under Q (in the open subset $x_0 > 0$), which is embedded. Then for $x \in Q$, we know $T_x H_k^n \subseteq T_x L^{n+1} = L^{n+1}$ is the kernel of dQ_x . We compute Q 's differential:

$$dQ_x = (-2x_0, 2x_1, \dots, 2x_n),$$

so

$$T_x H_k^n = \ker dQ_x = (-x_0, x_1, \dots, x_n)^\perp,$$

where $\bullet^\perp$ is the orthogonal complement relative to the usual Euclidean metric. Notice though that

$$(-x_0, x_1, \dots, x_n) \cdot (y_0, \dots, y_n) = -x_0 y_0 + \sum_{i>0} x_i y_i = Q(x, y).$$

Thus

$$\ker dQ_x = \{y \in L^{n+1} \mid (x, y) = 0\},$$

and since $\eta = x/r$, $(x, y) = 0$ iff $(\eta, y) = 0$. Thus η indeed is normal to $T_x H_k^n$.

Further notice that

$$(\eta, \eta) = \frac{1}{r^2} Q(x) = -1.$$

If we extend η to an orthogonal basis of $(\bullet, \bullet)$ we obtain new vectors $b_1, \dots, b_n$ such that $(\eta, b_i) = 0$, and (b_i, b_j) is diagonal. Since the signature of a bilinear form is invariant, we must have that $(b_i, b_j) = \delta_{ij}$ as desired. Then $b_1, \dots, b_n$ forms a basis for $T_x H_k^n$ as they are orthogonal to η , and then $(\bullet, \bullet)$ on $T_x H_k^n$ must be positive definite, by linearity (b_i form a positive orthogonal basis). Thus Q forms a Riemannian metric on H_k^n as desired.

- (2) The line between $p \in H_{-1}^n$ and p_0 is given by $tp + (1-t)p_0$, which intersects D^n when

$$tx_0 + (t-1) = 0 \iff t = \frac{1}{1+x_0}.$$

Thus

$$u_k = \frac{x_k}{1+x_0}.$$

For $(0, u_1, \dots, u_n) \in D^n$, let $U = \sum_i u_i^2$. In our case

$$U = \sum_i \frac{x_i^2}{(1+x_0)^2} = \frac{1}{(1+x_0)^2} \sum_i x_i^2,$$

and since $x \in H_{-1}^n$, $\sum_i x_i^2 = x_0^2 - 1$ so

$$U = \frac{x_0^2 - 1}{(1+x_0)^2} = \frac{x_0 - 1}{x_0 + 1}, \quad 1 - U = \frac{2}{1+x_0}.$$

And then

$$\frac{2u_k}{1-U} = \frac{\frac{2x_k}{1+x_0}}{\frac{2}{1+x_0}} = x_k$$

and

$$\frac{2}{1-U} - 1 = \frac{2}{\frac{2}{1+x_0}} - 1 = x_0$$

as desired.

Now for $X^i \partial / \partial x^i \in TH_k^n$, recall from the previous point that we have $Q(X, x) = 0$, so

$$X^i x_i = X^0 x_0.$$

Now we show that for $X^i \partial / \partial x^i, Y^j \frac{\partial}{\partial x^j} \in TH_k^n \subseteq TL^{n+1}$,

$$g\left(df\left(X^i \frac{\partial}{\partial x^i}\right), df\left(Y^j \frac{\partial}{\partial x^j}\right)\right) = \left(X^i \frac{\partial}{\partial x^i}, Y^j \frac{\partial}{\partial x^j}\right),$$

i.e. that f is an isometry. Since f can be extended to an open subset of L^{n+1} to map to L^{n+1} , we use linearity to compute $df(X^i \partial / \partial x^i)$:

$$df\left(\frac{\partial}{\partial x^i}\right) = \frac{\partial u^j}{\partial x^i} \frac{\partial}{\partial u^j}.$$

Now,

$$\begin{aligned} \frac{\partial u^j}{\partial x^j} &= \frac{1}{x_0 + 1}, \\ \frac{\partial u^j}{\partial x^i} &= 0, \quad \text{for } i \neq j, 0, \\ \frac{\partial u^j}{\partial x^0} &= -\frac{x_i}{(x_0 + 1)^2}. \end{aligned}$$

Thus

$$\begin{aligned} df \frac{\partial}{\partial x^i} &= \frac{1}{x_0 + 1} \frac{\partial}{\partial u^i}, \quad \text{for } i \neq 0, \\ df \frac{\partial}{\partial x^0} &= -\frac{x_i}{(x_0 + 1)^2} \frac{\partial}{\partial u^i}. \end{aligned}$$

So

$$\begin{aligned} g\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^j}\right) &= \frac{1}{(1 + x_0)^2} g_{ij} = \delta_{ij} \frac{1}{(1 + x_0)^2} \frac{4}{(1 - U)^2} = \delta_{ij}, \\ g\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^0}\right) &= -\frac{x_i}{(1 + x_0)^3} g_{ii} = -\frac{x_i}{(1 + x_0)^3} \frac{4}{(1 - U)^2} = -\frac{x_i}{1 + x_0}, \\ g\left(df \frac{\partial}{\partial x^0}, df \frac{\partial}{\partial x^0}\right) &= \frac{x_i^2}{(1 + x_0)^4} g_{ii} = \frac{x_i^2}{(1 + x_0)^4} \frac{4}{(1 - U)^2} = \frac{x_0 - 1}{x_0 + 1}. \end{aligned}$$

And finally we get

$$\begin{aligned} g\left(X^i df \frac{\partial}{\partial x^i}, Y^j df \frac{\partial}{\partial x^j}\right) &= X^i Y^j g\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^j}\right) = \\ &= \sum_{i \geq 1} X^i Y^i - \sum_{j \geq 1} \frac{x_j}{1 + x_0} X^0 Y^j - \sum_{i \geq 1} \frac{x_i}{1 + x_0} X^i Y^0 + X^0 Y^0 \frac{x_0 - 1}{x_0 + 1} \end{aligned}$$

Since

$$\left(X^i \frac{\partial}{\partial x^i}, Y^j \frac{\partial}{\partial x^j}\right) = -X^0 Y^0 + \sum_i X^i Y^i,$$

we want to show that

$$X^0 Y^0 = X^0 \sum_j \frac{x_j}{1 + x_0} Y^j + Y^0 \sum_i \frac{x_i}{1 + x_0} X^i - X^0 Y^0 \frac{x_0 - 1}{x_0 + 1}.$$

Recall that $X^i x_i = X^0 x_0$, and thus the right-hand side is equal to

$$2X^0 Y^0 \frac{x_0}{x_0 + 1} - X^0 Y^0 \frac{x_0 - 1}{x_0 + 1} = X^0 Y^0$$

as desired.

- (3) Let $\bar{g}$ denote the metric on D^n , so $\bar{g}_{ij} = 4\frac{\delta_{ij}}{(1-U)^2}$. The idea here is the same as the other points: compute the Jacobian of f , and show that it maps isometrically. Here we can focus only on the coordinate frame $\partial/\partial u^i$ since D^n is open, and thus the coordinate frame spans its tangent space. Let $f(u_1, \dots, u_n) = (x_1, \dots, x_n)$, then we have

$$df \frac{\partial}{\partial u^i} = \frac{\partial x^j}{\partial u^i} \frac{\partial}{\partial x^j},$$

and we want to show that

$$\bar{g}_{ij} = g\left(df \frac{\partial}{\partial u^i}, df \frac{\partial}{\partial u^j}\right) = \frac{\partial x^k}{\partial u^i} \frac{\partial x^\ell}{\partial u^j} g\left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell}\right) = \sum_k \frac{\partial x^k}{\partial u^i} \frac{\partial x^k}{\partial u^j} \frac{1}{x_n^2}.$$

The computations here became too messy for me to do, but that's the idea.

- (4) Let $\bar{g}$ denote the metric on H_{-1}^n , so we want to show that for $X = X^i \partial/\partial x^i, Y = Y^j \partial/\partial x^j \in T_x H_k^n$,

$$-X^0 Y^0 + \sum_i X^i Y^i = \bar{g}(X, Y) = g(dh(X), dh(Y)).$$

Of course we need to compute $dh \frac{\partial}{\partial x^i}$,

$$dh \frac{\partial}{\partial x^i} = \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j},$$

so

$$\begin{aligned} dh \frac{\partial}{\partial x^i} &= \frac{1}{x_0} \frac{\partial}{\partial y^i} \\ dh \frac{\partial}{\partial x^0} &= - \sum_i \frac{x_i}{x_0^2} \frac{\partial}{\partial y^i}. \end{aligned}$$

And so

$$\begin{aligned} g\left(dh \frac{\partial}{\partial x^i}, dh \frac{\partial}{\partial x^j}\right) &= \frac{1}{x_0^2} g_{ij}, \\ g\left(dh \frac{\partial}{\partial x^i}, dh \frac{\partial}{\partial x^0}\right) &= - \sum_j \frac{x_j}{x_0^3} g_{ij}, \\ g\left(dh \frac{\partial}{\partial x^0}, dh \frac{\partial}{\partial x^0}\right) &= \sum_{i,j} \frac{x_i x_j}{x_0^4} g_{ij}. \end{aligned}$$

And thus

$$g(dhX, dhY) = X^i Y^j g\left(dh \frac{\partial}{\partial x^i}, dh \frac{\partial}{\partial x^j}\right) = X^i Y^j \frac{1}{x_0^2} g_{ij} - X^0 Y^i \frac{x_j}{x_0^3} g_{ij} - Y^0 X^i \frac{x_i}{x_0^3} g_{ij} + X^0 Y^0 \frac{x_i x_j}{x_0^4} g_{ij}.$$

Note that since $x \in H_k^n$,

$$|y|^2 = \sum_i y_i^2 = \sum_i \frac{x_i^2}{x_0^2} = \frac{x_0^2 - 1}{x_0^2} \implies 1 - |y|^2 = \frac{1}{x_0^2}.$$

Thus

$$g_{ij} = \delta_{ij} x_0^2 + x_i x_j x_0^2.$$

And so

$$g(dhX, dhY) = (\delta_{ij} + x_i x_j) \left(X^i Y^j - X^0 Y^i \frac{x_j}{x_0} - Y^0 X^i \frac{x_j}{x_0} + X^0 Y^0 \frac{x_i x_j}{x_0^2} \right).$$

So we want to show that

$$\begin{aligned}
 (*) \quad -X^0 Y^0 &= -X^0 \sum_i Y^i \frac{x_i}{x_0} - Y^0 \sum_i X^i \frac{x_i}{x_0} + X^0 Y^0 \sum_i \frac{x_i^2}{x_0^2} \\
 &\quad + \sum_{i,j} x_i x_j \left(X^i Y^j - X^0 Y^i \frac{x_j}{x_0} - Y^0 X^i \frac{x_j}{x_0} + X^0 Y^0 \frac{x_i x_j}{x_0^2} \right).
 \end{aligned}$$

Note that

$$\sum_{i,j} Y^i x_i x_j^2 = Y^0 x_0 \sum_j x_j^2 = Y^0 x_0 (x_0^2 - 1).$$

Thus the right-hand side of (*) is equal to

$$-X^0 Y^0 - X^0 Y^0 + X^0 Y^0 \frac{x_0^2 - 1}{x_0^2} + x_0^2 X^0 Y^0 - (x_0^2 - 1) X^0 Y^0 - (x_0^2 - 1) X^0 Y^0 + X^0 Y^0 \frac{(x_0^2 - 1)^2}{x_0^2}.$$

All this just simplifies to give $-X^0 Y^0$, as desired.